

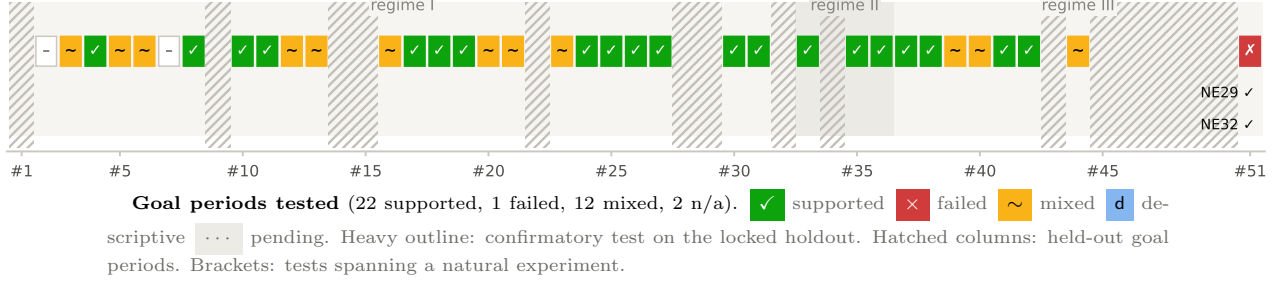
H89: Price equation for village culture

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 1.5/5

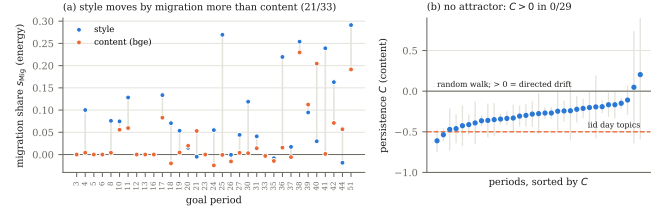
Bottom line. Content and conventions move by transmission and style partly by migration, but selection is undetectable and the remainder is day-topic swing, not a village attractor.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

The test splits the day-to-day change of the village's mean trait into selection, transmission and migration with a Price equation. Traits are content (DQ5 vectors, style removed), style (H13's 20 features) and convention use. Fitness counts in-cone adopters from the context ledger. Over 33 non-holdout periods (245 day transitions), stayers' own change carries the content change: median transmission share 0.98 (33/33). Selection carries 0.01 and never beats its permutation null. Style moves by migration more than content in 21 of 26 periods with turnover, but its migration share is only 0.04 (max 0.29). The kickoff direction carries 0.04, so the kill does not fire. The remainder swings day to day with no attractor (persistence $C > 0$ in 0/29). NE29 and NE32 pass. #51's eleven joins do not move its net style. The holdout script is not run.



Observables. (a) Migration share of the day-to-day change per goal period, for style (blue) and content (orange). Style sits above content in 21 of the 26 periods with any turnover. Seven periods have no turnover, so both shares are 0. (b) Persistence C of the non-kickoff content change, with 95% jackknife CIs. Independent daily topics give -0.5 . A village attractor needs $C > 0$, which no period reaches.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor of collective culture: within a goal period, the village's mean content follows what the stayers do each day, not who is present and not who is influential. Read a content shift as a day field acting on everyone. Influential agents do not pull the mean ($|s_{\text{sel}}| \approx 0.01$). Style is the roster-sensitive trait: it gets about ten times content's migration share (0.044 against 0.004). A style change on a day with a join or exit is therefore mostly composition. The split cannot tell copying from a common field at day resolution. Do not use it as evidence that agents copy each other. It is a bookkeeping identity with honest error bars. It is not a lever.

PHYSICAL MODEL

Cultural Price equation (model 13) with in-cone parentage. A trait $z_i(d)$ is agent i 's day- d mean. Generations are consecutive active days. Stayers S , leavers E and entrants I define the open population. Stayer j inherits $1 - \lambda_j$ from itself and λ_j from its in-cone sources. λ_j is the share of its first uses that are in-cone adoptions. Fitness w_i sums the parentage weights, $\bar{w} = 1$. Shares s_X are cross-fitted between message halves A and B (symmetrized). C is the cross-fitted cosine of consecutive non-kickoff changes.

$$\Delta \bar{z} = \text{Sel} + \text{Trans} + \text{Mig}, \quad \text{Sel} = \text{Cov}_S(w, z), \quad \text{Trans} = \text{E}_S(w \Delta z)$$

$$\text{Mig} = \frac{1}{|A_d|} \sum_I (z_j - \bar{z}_S) - \frac{1}{|A_d|} \sum_E (z_i - \bar{z}_S)$$

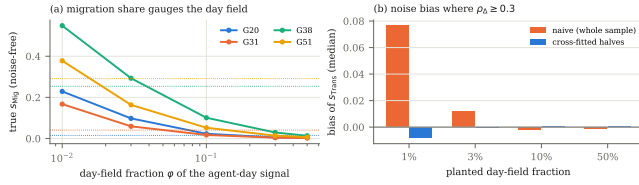
$$w_i = \sum_{j \in S} \alpha_{ji}, \quad \alpha_{jj} = 1 - \lambda_j$$

$$s_X = \sum_t \langle X_t^A, \Delta_t^B \rangle_{\text{sym}} / \sum_t \langle \Delta_t^A, \Delta_t^B \rangle$$

H89-price-equation-culture · generated 2026-10-04 13:36 by in-fra/summaries/build_summaries.py · card hypotheses/H89-price-equation-culture/README.md · ratings: H89 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1a style migration > content	21/33 (p 0.08); ties excluded 21/26	fail (narrow)
P1b migration largest for style	0/33; median s_{Mig} 0.044	fail
P2 content: transmission ≥ 0.5	33/33, both models; median 0.98	pass
P3 $ s_{\text{Sel}} \leq 0.2$	33/33 for each trait; median 0.01–0.03	pass
P4 selection beyond null	1–3 of 33 per trait (chance 1.7)	descr.
P5 kill (R CI $\ni 0$)	R 0.92, CI > 0 in 33/33; kickoff 0.04	not killed
P6 attractor $C > 0$	0/29; median -0.28	fail
N1 NE29 retirement	style 0.16/0.03 vs content 0.05/0.00	pass
N2 NE32 newcomers	entry style 0.08 vs 0.02; content cos +0.26	pass
N3 #51 growth	style roster share -0.14 [$-0.34, +0.05$]	fail



Second figure. Synthetic data at real counts. (a) The noise-free migration share falls steeply with the day-field fraction ϕ . Dotted lines are the observed style shares. They imply $\phi \approx 0.07$ for style and 0.32 for content. (b) Without cross-fitting, sampling noise inflates transmission by up to 0.08. Cross-fitting removes this bias.

GOAL PERIODS IN DETAIL

G03 ~ mixed
G04 ✓ supported
G05 ~ mixed
G06 ~ mixed
G08 ✓ supported
G10 ✓ supported
G11 ✓ supported
G12 ~ mixed
G13 ~ mixed
G16 ~ mixed

+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

A day is a coarse generation. Presence turnover makes up most of the migration share; roster events are rare inside periods. The fitness spread is small ($\bar{\lambda} = 0.28$), so selection in a 20–32-dimensional trait is bounded by design, and no valid test has power for it. The synthetic copy scenario gives the same shares as a pure field. The unread placebo has a median of 8 adoptions per period. Cumulative shares are not causal: with persistent fitness, selection and transmission accumulate and cancel. N2 rests on one newcomer across the merge, and the gte model does not reproduce its content cosine.

ROUND 2

Measure heritability directly: regress each adopter's next-day trait on its in-cone parents beyond the day mean, with unread parents as the placebo. Run the split on single axes and across goal switches with roster change.

- **H89-R1** (separate copying from fields): a parent–offspring regression per transition: does j 's day- d ' trait move toward its in-cone parents' day- d traits beyond the day mean, with unread parents as the placebo?
- **H89-R2** (per-axis Price): run the partition on single interpretable axes (the goal_fields directions, H13 style features one at a time, single conventions) so selection is not diluted by 20–32 dimensions of day field.
- **H89-R3** (cross-boundary Price, exception (c)): run the split across goal switches with roster changes, where migration is large.
- **H89-R4** (a valid selection null): permute agent labels for the whole period (fitness trajectories kept), so the persistent-fitness inflation of the magnitude test goes away.